

Chapter-2

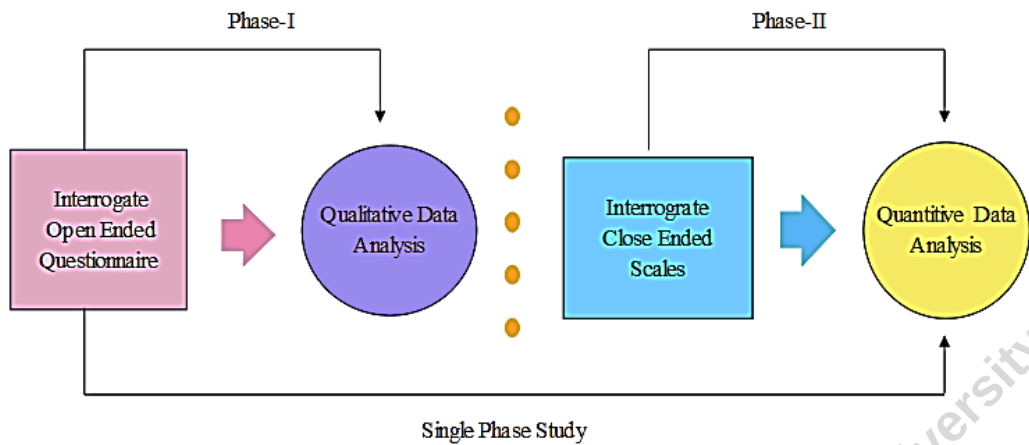
Method

It is considered as vital step in research because the research investigation cannot be obtained unless it is carried out in a systematic and planned way. Methodology is the rationale for using scientific procedures to study phenomena (Hedges & Cooper, 2009; Mouno & Marais, 1993) . Therefore, it is a kind of decision-making process in which the investigator has to select the most appropriate research design, suitable sampling technique and measuring tools. Thus, the objectivity of scientific investigation depends upon the accuracy of the research methodology adopted by the researcher.

In the present study “guilt, shame and remorse related to appraisal: exploring social emotions” mainly focused on two aims and are i.e., 1) Explore the situations that generate the social emotions of guilt, shame and remorse, and 2) Evaluate the emotional experiences of guilt, shame and remorse with the levels of appraisal. Both quantitative and qualitative data were collected in a single phase. However, the division of phase is explained into two separate parts such as phase I-qualitative analysis for fulfillment of aim 1 and phase II- quantitative analysis for targeting of aim 2.

2.1 Description of Research Design

Mixed method research are philosophically unpinned model of inquiry combining both quantitative and qualitative methods of research so that the evidence may be mixed and knowledge is increased in a more meaningful manner than either model could achieve alone (Creswell & Plano Clark, 2007). Therefore, in order to fulfill the respected aims of the present study, it is necessary to design and plan a right research design and strategy. Consequently, the design contains diverse types of methods and tools that are relevant to accomplish the research. But, it's importance to utilize appropriate tools and research methods that are highly reliable and valid. In this way, the current research strategy of this thesis began by collecting the qualitative data first and then moves towards the quantitative data at the same line (I.e., by focusing on single phase study). Therefore, the parallel mixed method research design was used in the present study. (See Fig; 2.1)

Figure 2.1*Mixed Method Research Design (Schema)*

A parallel mixed method design is single phase research approach that involves collecting both quantitative and qualitative data, analyzing them separately, and then combining the findings to provide a more comprehensive understanding of the research problem. The main characteristic of this design is that the quantitative and qualitative data are collected and analyzed concurrently, and the results are integrated during the interpretation phase.

Firstly, in phase-I (qualitative design) the exploratory phenomenological approach was used to answer the questions of aim 1 and it's an approach of inquiry coming from philosophy and psychology view in which the researcher describes the living experiences of individuals about the phenomena or event as described by participants. In this approach researcher explored the phenomenon of individual experiences of guilt, shame and remorse while appraising the particular phenomenon or situation. The data was collected with the help of certain open ended and semi-structured questions related to situation or phenomenon encountered by the participants. Because, the design has strong philosophical foundations and help the researchers to discover different typologies situations related to social emotions (guilt, shame and remorse) that individuals face in everyday life (Giorgi, 2009; Moustakas, 1994).

Secondly, the phase-II (quantitative design) attempts to achieve the objectives of aim 2. The survey approach was carried as an extension of phenomenological approach focused on quantitative side. In this phase of research, the researcher did not have greater control over manipulation of factors. The researcher highlights the bodily

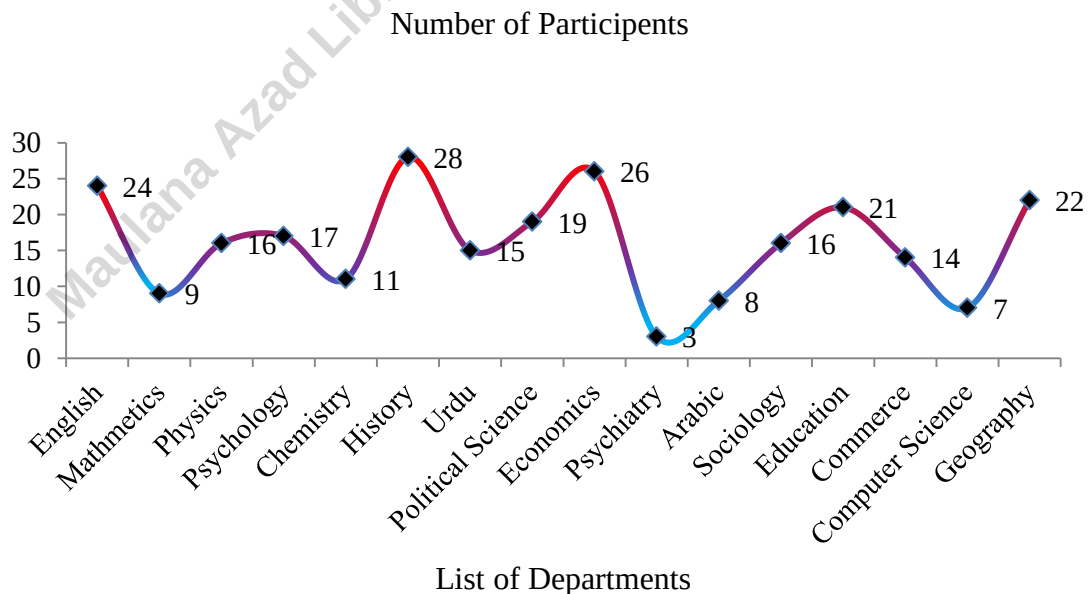
symptoms, expressive reactions, verbal reactions and psychological consequences experienced by the participants while describing the particular phenomenon or situation happen in his life.

2.2 Participants

In the present study the primary data was collected and the target population of the study was Post-graduate and higher students of Aligarh Muslim University preferably belonging from the states of Jammu and Kashmir and Uttar-Pradesh. All the participants were registered in different faculties (such as Arts, Social science, Commerce, Sciences, etc.) of Aligarh Muslim University and participated voluntarily in the current study (See fig; 2.2). The place of Jammu and Kashmir is divided into three regions (Jammu, Kashmir and Ladakh) based on distinct geographical regions and social features (Chandramouli & General, 2011). The valley is truly a dream of beauty set in the heart of Himalayas. On the other hand, the state of Uttar Pradesh presented at Centre of the country with large population and hot weather and humid conditions. Both the states are apart with their cultural, customs and tradition patterns.

Figure 2.2

Postgraduate and Ph.D. Students Participated From Different Faculties of Aligarh Muslim University From the State of Uttar Pradesh and Jammu & Kashmir



The total numbers of 264 adult participants were approached for the data collection out of which eight participants were dropped for missing information. Therefore, the final sample comprised 256 participants in the study. The age ranged from 18-30 years ($M_{\text{age}}=25.01$ years, $SD=1.759$) belonging to the states of Jammu and Kashmir ($n=115$) and Uttar-Pradesh ($n=141$). The reason for considering broader age range was given by Alwin (1994) in the persistence model. The researcher highlighted a uniform high level of stability depicted over each segment of the adult life cycle. In this model of expected magnitude, on a cycle range from 0 to 1, over each such 8-year period depicts a high level of stability that was uniform with the respected age. Traits for which such patterns were observed would seemingly represent a very high level of stability in individual differences across the entire adult life span. For many human characteristics, such as cognitive and intellectual abilities, various aspects of identities, and some personality traits, there was a strong empirical basis for assuming high levels of stability after adolescence and early adulthood (Alwin, 1994). The gender composition of participants was 40.2% male and 59.8% female and their participation was entirely voluntary and they did not received any type of incentive for participating in the study. (See Table 2.1)

Selecting the right *sample size* the G*power analysis was conducted because too small of a sample size can produce inconclusive or unreliable results and too large of a sample size may lead to unnecessary wasting of resources. Therefore, a statistical power analysis was conducted and using G*Power version 3.1.9.4 (Faul et al., 2009) to determine the correct sample size required to test the study hypothesis. Results indicated the required sample size to achieve 95% power for detecting a medium effect, at a significance criterion of $\alpha = .05$, was $N = 172$ for MONAVA, Chi Square and Independent Sample t-test. Thus, the obtained sample size of $N = 172$ is adequate to test the study hypothesis. The G* power statistical data sheet is attached as Appendix-A.

Table 2.1*Socio-Demographic Distribution of Participants (N=256)*

Socio-demographics	N	%
State		
Jammu & Kashmir	115	44.9
Uttar Pradesh	141	55.1
Gender		
Male	103	40.2
Female	153	59.8
Residential status		
Urban	143	44.1
Rural	113	55.9
Educational Qualification		
Doctorate (Ph.D.)	153	59.8
Post Graduate	103	40.2
Language		
Kashmiri	115	44.9
Hindi	141	55.1
Fathers Occupation		
Public	108	57.8
Private	148	42.2
Income Source		
Private	100	39.9
Public	105	41.0
Both	51	19.9

Note. The mean $_{Age}$ = 25.01, SD = 1.75

For *sampling* the researcher selected criterion-based purposive sampling to select the participants. In this sampling strategy the sampling was carried out by the researcher with the assumption that the sample is representative of the characteristics of the population. Researchers preferred to use criterion-based purposive sampling strategy in qualitative research because particular individuals are chosen with characteristics relevant to the study that are thought will be most informative. In this way, the current study relies on criterion based purposive sampling strategy and

certain criterion was used like age, state, gender and urban and rural to match the subjects. Further, researchers revealed that criteria strategy is commonly used in qualitative researchers as a means to select on a characteristic or group of characteristics known to vary within a population (Teddie & Yu, 2007). Thus, the strategy helps to select a more comparative interest of the sample and help in findings to representative the large set of population (Krefting, 1991).

2.3 Inclusion Criteria

- The voluntary willingness of participants.
- Both male and female participants were included in the study.
- Participants within the age group of 18-30 years.
- Participants belonged to Uttar-Pradesh and Jammu & Kashmir states.
- Both rural and urban participants were included in the study.

2.4 Exclusion Criteria

- The participants who were not innate and grow in their home states.
- The overseas and state migration participants belonging from respected states were excluded.
- Participants who have experienced any traumatic experience.

2.5 Psychometric Measures

2.5.1 Socio-Demographic Data Sheet

Primarily starting from the consent form and then collecting background information of the participants. The consent form gives security and authority to the participant to leave the study if he feels uncomfortable by sharing their confidential information. The socio-demographic data sheet was also designed to collect basic background information of the population to the study. The sheet consists of information pertaining to age, gender, education qualification, schooling, state, residential status, language, father's occupation, source of income were considered to ensure homogeneity in the sample. For confidentiality, the participants were requested not to write their name or any signature on the sheets. Both consent form and socio-demographic data sheet is attached are *Appendix-B & C*.

2.5.2 Emotional Experience Questionnaire

The EEQ (Emotional Experience Questionnaire) was used in the current study. The EEQ carried out in 37 countries and validated by the international survey on emotion antecedents (Scherer & Wallbott, 1994). Moreover, the meaning of each emotion in their native language was added to follow the cultural norms such as in *Hindi and Urdu* languages the **shame** emotion are called **Lajja** / लज्जा and شرمند .Wherein Kashmiri language it's called **Man-dash**. Similarly, the **guilt** emotion in *Hindi and Urdu languages* are called **Aparaadhbodh** / अपराधबोध and حرم and in Kashmiri language it's called **Aa-tur** and finally, the **remorse** emotion in *Hindi and Urdu* are called **Pashchaatap** / अपराधबोध and پچھتاوا . Wherein Kashmiri language it's called **Afsoos/ Pash-taw** . Further, the graphical rating scale added a visual element to convey the varying emotions and enhancing the understanding of their experiences. The pictures are tranquil form (IAPD) Indian Affective Picture Data Base (Sharma & Bhushan, 2019) and (IDBE) Indian Dataset of Basic Emotions (Bhushan, 2015), so that the participants can easily understand the notion of emotions. Lastly, add items of psychological consequences in each set of social emotion. However, after doing the pilot study the research observed that there was no effect on previous reliability by doing certain changes and instead the new (Chronbech α) reliability increases from .811 to .824 (proved through pilot study).

The instructions given the participant were to recall the experienced situation of guilt, shame and remorse with regard to the level of appraisal (Low, Moderate and High) and report the reminiscence appropriate behavior and reactions which were happened during that event and mark against the statement situations accordingly. For each social emotion of guilt, shame and remorse the questionnaire mainly distinguished into five parts: (I) Describing the emotional situation I) Graphical rating of situation by appraisal levels (Low, Moderate and High) (III) Subjective feeling state (regarding occurrence, longevity and frequency and intensity) (IV) Physiological reactions (bodily symptoms, non-verbal and expressive reactions) (V) Psychological concerns.

Further, feedback of the participants was recoded in order to use in analysis and make sound decision regarding to social emotion. The Emotional Experience Questionnaire (EEQ) is attached as *Appendix- D*.

2.5.2.1 Situation Description. Participant's shared brief accounts (30-50 words) of guilt, shame, and remorse emotion experience situations by using a graphical rating scale to express the levels of appraisal and intensity of these social emotions.

2.5.2.2 Subjective Feeling State. The participant also has to mention the subjective feeling state regarding to social emotions i.e., when did the situation occur, how many days, months or years back it happen, and what was the frequency of reoccurrence of the situation.

2.5.2.3 Physiological Symptoms. The ergo tropic and trophotropic systems (Anolli & Pascucci; 2005), made by Gellhorn and Loofburrow (1963) was used. (a) Trophotropic symptoms: lump in throat, crying/sobbing and stomach troubles; (b) Ergo tropic symptoms: heart beating faster, change in breathing, perspiring/moist hands and muscles tensing/ trembling; (c) Felt temperature: feeling warm/pleasant, feeling cold/shivering and cheeks burning/ feeling hot (0 being given when no symptom was reported).

In expressive behavior, the four variables are composite: (a) Behavior Movements: moving away from people and things, moving toward them (b) Non-verbal behavior: crying/sobbing, smiling/ Laughing, other facial expression changes, yelling/ screaming, changes in gesturing and other speech variations (c) Paralinguistic behavior: speech disturbances, speech melody change and speech tempo change (d) Verbal behavior, for which respondents were asked to say what they were doing (Scherer & Wallbott, 1994).

2.5.2.4 Psychological Consequences. The participants were asked to mention the experiences of psychological consequences that were happen during the guilt, shame and remorse situation. The psychological consequences variables composite in (I) *Guilt*: Feeling bad about my behavior, worried what to do next, thinking my mistakes for several hours, difficulty to enjoy and feel alone, feeling inadequate, fear of punishment, work hard to get rid of bad thoughts, feeling repentance, getting frustrated and angry and treating myself badly. (II) *Shame*: feelings of low self-esteem, demotivated and blushing, negative thought outbreak, feeling like other takes advantage, avoid being central attention, want withdrawal the situation, feeling dishonorable, feeling sad and uncomfortable, keeping mistakes hidden to avoid

embarrassment and feeling anxious and stressful. (III) *Remorse*: feeling of bitterness, feeling of sorrow, disappointment and regret of behavior (0 being given when no symptom was reported).

2.6 Procedure

Before collecting data the permission was obtained from the Research Advisory Committee (RAC) and Board of Studies (BoS) from the Department of Psychology, Aligarh Muslim University, Aligarh (India). Formerly taking part in the research, the researcher and participants introduced to each other and a proper rapport was build. The participants were asked for their consent to take a part in the study and then a short explanation to the research was given to him. Further, the participants were made comfortable and assured about the confidentiality of data through consent form. Apart from this participants were also given freedom to withdraw or discontinue any time during the study, if they feel discomfort for sharing their information. After debriefing the purpose of the study, the socio-demographic sheet and set questionnaires were given and participants were asked to read the information carefully provided in the given sheets. However, if the participant feel distressed while recalling experiences of guilt, shame and remorse, he or she can leave the testing session and point out their problem to researcher. The researcher provides the necessary help in managing distress and by providing the dialectic counseling therapy instruction to relief immediate distress level (Griffin et al., 2023). After collecting the data the participant was acknowledge and thanked for their participation.

2.7 Data Analysis Methods

2.7.1 Phase I: Qualitative Data

Text Mining and Content analysis was used to measure aspects of content (Eriyanto, 2011). The procedure was to measure the features of content and present them quantitatively (Gandasari & Dwidienawaiti, 2020). Content analysis is mainly used to analyses opinion of individuals through interviews, written documents, print and electronic media, such as newspapers, radio, television and film (Eriyanto, 2011). Therefore, in the present study the text mining and content analysis are used on written document experiences of guilt, shame and remorse.

Initially, the researcher entered the data of open-ended text questions of guilt, shame and remorse experience and then text mining are used to identify the nodes

with the help of natural language processing (NLP) by using Rapid Miner software. Later on the content analysis was used to relate the nodes with the respected situation experienced by individual. The object of the content analysis was to examine written texts of guilt, shame and remorse experienced by the participant in his life.

The Rapid Miner 9.9 version software aim was to generate the suitable codes from the past experiences of guilt, shame and remorse of the participant's. The codes was generated by applying the sequential natural language processing (NLP) steps such as inserting and prepare the data (Term Frequency-Inverse Document Frequency (TF-IDF)), parts of speech tagging, tokenizing, stop wording, filtering tokens, gram (one phrases), stemming and finally nodes. In this way, the appropriate nodes are generated and are visualized with the illustration of grid graph and further analysis was taken through NVivo software.

For this, the NVivo +12 version software tool was used to categories the content associated with nodes draw form rapid miner that determines the social emotions (guilt, shame and remorse) experiences of individuals. NVivo is data analysis software that can import texts, audios, videos, images, news articles, spreadsheets, online surveys, web content and social media from various sources into a simple, intuitive interface (McNiff, 2016; Paulus & Lester, 2020). Further, the software help in organizing, classify, sort, analyzing and examine relationships in the data.

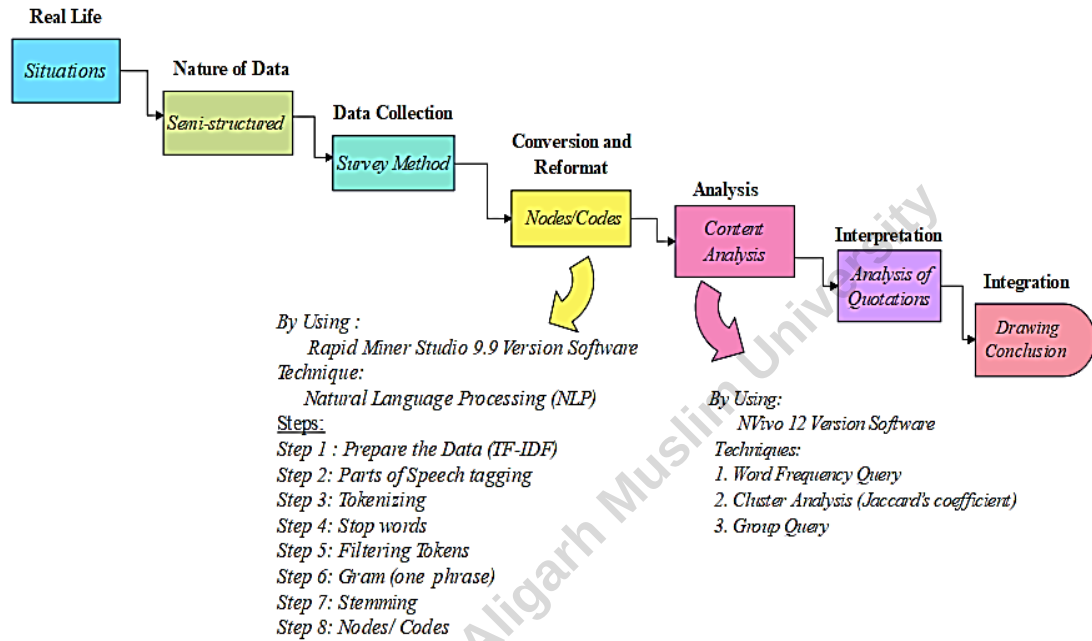
The steps in content analysis using NVivo were (1) *import*: importing text documents into NVivo, using transcription; (2) *organize*: organizing codes, coding texts and creating codes; applying coding stripes and highlights; using cases with classifications and attributes; and (3) *explore*: exploring queries; applying code and word frequency query; and illustrating with visualizations by cluster analysis are shown. (See Fig; 2.3)

The cluster analysis use to visualize patterns in nodes that share similar words, similar attribute values, or are coded similarly by nodes. Cluster analysis diagrams provide a graphical representation of nodes to make it easy to see similarities and differences. The nodes in the cluster analysis diagram that appear close together are more similar than those that are far apart. This can be also checked with the help of *Jaccard's coefficient* with the range of 0 to 1 (i.e., 0 = indicates least similarity and 1=

indicates most similarity). Thus, the study used grid representation for identification of most situation word frequency count; table describes the quantitative result and cluster analysis describes the similarity of nodes.

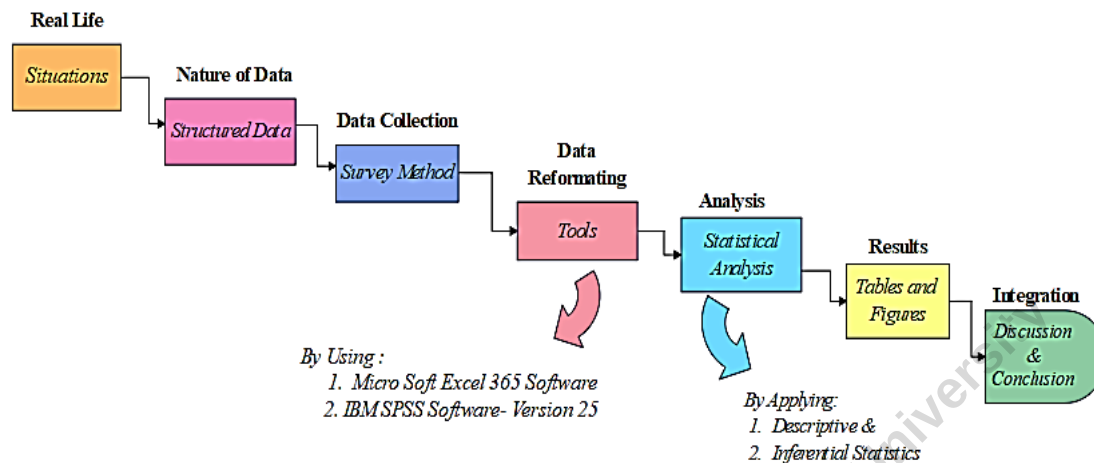
Figure 2.3

Steps of Qualitative Analysis: Methods and Tools Used



2.7.2 Phase II: Quantitative Data

The data related to phase II (Quantitative data) of research was entered into Microsoft excel. The analysis was carried out by using IBM SPSS 25 version. Before proceeding to the final analysis the basic assumptions of the data have been checked out i.e., the missing values, outliers, range and inflated SD of the data were examined. The appropriate statistical tests were employed to analysis the descriptive and inferential statistics to fulfill the aims and their associated objectives of the study. The results of the analysis were presented according to the APA 7th publication manual (Bradley et al., 2020) and the findings highlighted in the line of existing literature. (See Fig; 2.4)

Figure 2.4*Quantitative Analysis: Statistical Tests and Techniques Applied*

2.8 Research Ethics

The research was of the non-invasive kind and posed no substantial concern or damage to the participant. The present study followed the guidelines of American Psychological Association (APA) and Indian Council of Medical Research's (ICMR) principles for ethical research. Further, the current study also followed the rules and regulations of the respected intuition (Aligarh Muslim University) too. Likewise the study was first approved by the Board of Studies (BoS) and Committee of Advanced Study and Research (CASR). Furthermore, the permission to data collection was also taken from the RAC and head of the Department of Psychology before approaching the respondents.

2.9 Pilot Study

A pilot study was carried out before the main study to evaluate the viability of the research instruments and the efficacy of the research methodology for the intended audience. 118 adults from the states of Jammu & Kashmir and Uttar Pradesh were selected from the suggested population for the pilot project. The major goal of the pilot research was to evaluate, finalize the tool modification, and decide if the inclusion and exclusion criteria for the main study were adequate. Following the pilot study's findings certain modifications were made to the questionnaire such as adding Hindi, Kashmiri and Urdu language names of these emotions into scale. So that the

participants were easily understand the social emotions (guilt, shame and remorse) in their native language.

2.10 Publishing Research Findings

A necessary step in any research process is the publication of results. If a study's findings are not shared with those who stand to gain from it, it is not complete. Some of the present study's findings were published in prestigious journal in order to meet ethical and research standards.

Maulana Azad Library, Aligarh Muslim University

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